

Comparative Analysis of Competitiveness in European Football League and MLS

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Research Question

- What Key Team-level Performance Metrics Do European Football Leagues Demonstrate Greater Competitiveness Than MLS?

□ Foundational Assumption:

European Leagues are inherently more competitive than MLS

EU Leagues: Competitive = 1, MLS: Competitive = 0

Methodology



- **Data (Fbref):**

Team-level standings, fixtures, and performance stats from 2010–2024 across 5 major European leagues + MLS.

- **Feature Selection:**

Used RFECV to choose top predictive metrics

- **Modeling Approaches:**

Logistic Regression & Decision Tree Classifier

- **Cross-Validation:**

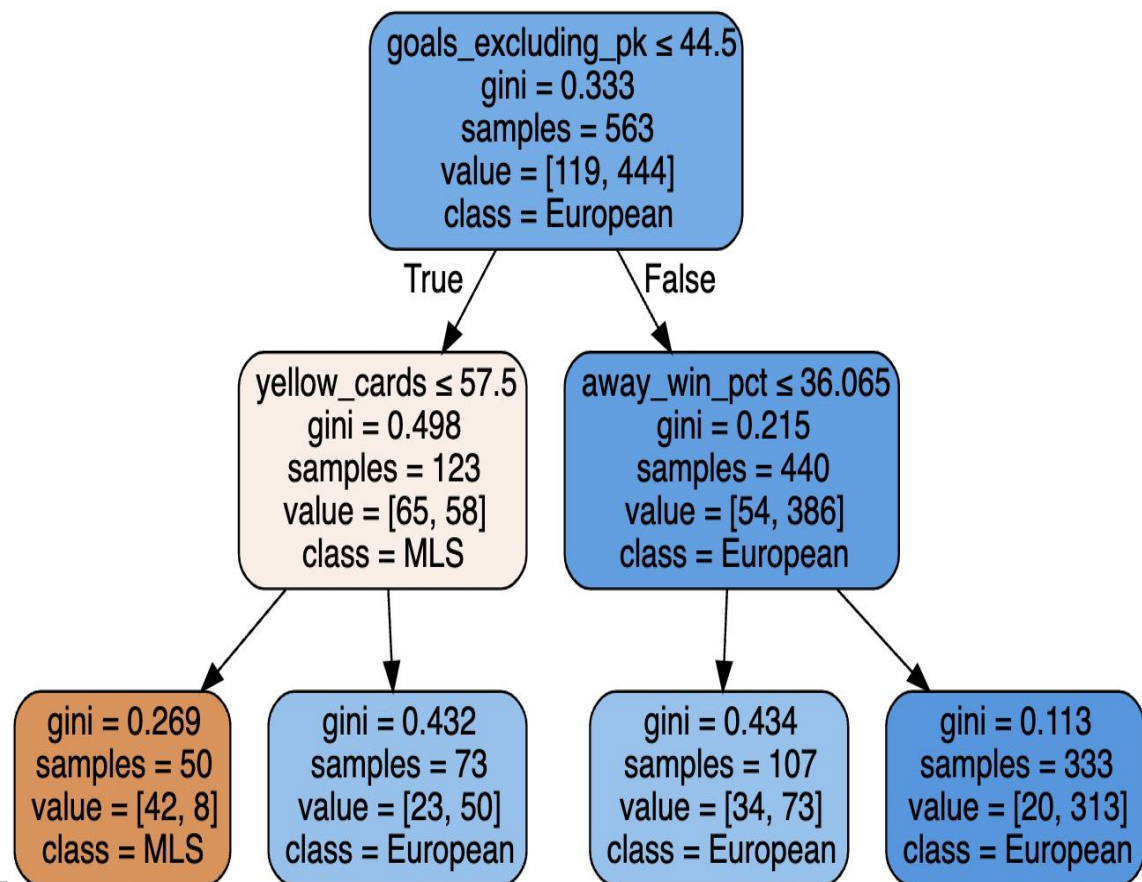
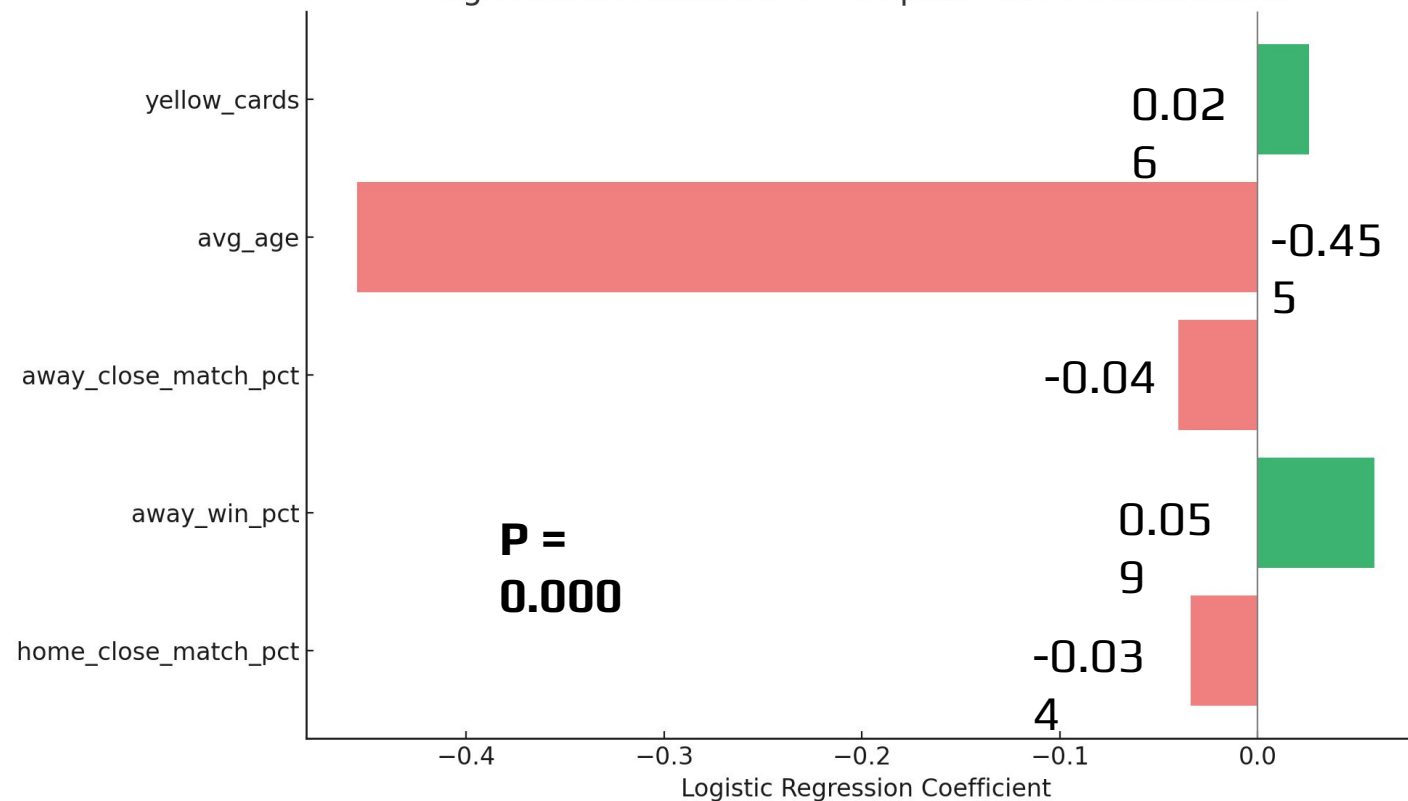
Stratified K-Fold CV (5-fold)

- **Two Datasets Modeled:**

- All teams across all seasons
 - Only qualified/high-performing teams (e.g., UEFA or conference teams)
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Results

Significant Predictors of European Team Classification



Takeaways & Discussions



European teams are more likely to:

- Have **higher away win percentages**
- Drag into less **close match** situations (both home & away)
- Older **squads**
- Accumulate **slightly more yellow cards**.

Discussion

- Based on Common Beliefs
- Maybe reveal **structural and stylistic differences**, but no results:
- **MLS teams** show more **parity and variability**
- **European teams** show **consistency, control, and tactical maturity**

Future Directions

- Try **unsupervised clustering** to explore natural groupings without assumptions.
- Introduce **more advanced metrics** (e.g., expected goals, possession zones).
- Analyze off-court metrics